

50 Coding Questions

1. Check Even or Odd

```
```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc = new Scanner(System.in);

 int n = sc.nextInt();

 System.out.println(n % 2 == 0 ? "Even" : "Odd");

 }

}

```
```

2. Check Prime Number

```
```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc = new Scanner(System.in);

 int n = sc.nextInt();

 boolean prime = true;

 if(n <= 1) prime = false;

 for(int i = 2; i <= Math.sqrt(n); i++){

 if(n % i == 0){

 prime = false;

 }

 }

 }

}

```
```

```
break;
}
}
System.out.println(prime ? "Prime" : "Not Prime");
}
}
...

```

3. Factorial of a Number

```
```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

long fact = 1;

for(int i = 1; i <= n; i++)

fact *= i;

System.out.println(fact);

}

}

...

```

### **## 4. Fibonacci Series (First N Terms)**

```
```java

```

```

import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int a = 0, b = 1;

for(int i = 1; i <= n; i++){

System.out.print(a + " ");

int c = a + b;

a = b;

b = c;

}

}

}

...

```

5. Reverse a Number

```

```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int rev = 0;

while(n != 0){

rev = rev * 10 + n % 10;

n /= 10;

```

```
}
System.out.println(rev);
}
}
...

```

### **## 6. Check Palindrome Number**

```
```java  
import java.util.*;  
  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int temp = n, rev = 0;  
        while(n != 0){  
            rev = rev * 10 + n % 10;  
            n /= 10;  
        }  
        System.out.println(temp == rev ? "Palindrome" : "Not Palindrome");  
    }  
}  
...  

```

7. Armstrong Number

```
```java
```

```

import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int temp = n, sum = 0;

while(n != 0){

int digit = n % 10;

sum += digit * digit * digit;

n /= 10;

}

System.out.println(temp == sum ? "Armstrong" : "Not Armstrong");

}

}

...

```

## ## 8. Sum of Digits

```

```java

import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int sum = 0;

while(n != 0){

sum += n % 10;

n /= 10;

```

```
}  
System.out.println(sum);  
}  
}  
``
```

9. Largest of Three Numbers

```
``java  
import java.util.*;  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();  
        int max = Math.max(a, Math.max(b, c));  
        System.out.println(max);  
    }  
}  
``
```

10. GCD of Two Numbers

```
``java  
import java.util.*;  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
        int a = sc.nextInt(), b = sc.nextInt();
```

```

while(b != 0){
int temp = b;
b = a % b;
a = temp;
}
System.out.println(a);
}
}
...

```

11. LCM of Two Numbers

```

```java
import java.util.*;

class Main{
public static void main(String[] args){
Scanner sc = new Scanner(System.in);
int a = sc.nextInt(), b = sc.nextInt();
int x = a, y = b;
while(y != 0){
int temp = y;
y = x % y;
x = temp;
}
int gcd = x;
int lcm = (a * b) / gcd;
System.out.println(lcm);
}
}

```

```
}
...

```

## **## 12. Check Leap Year**

```
```java  
import java.util.*;  
  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
  
        int year = sc.nextInt();  
  
        if((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))  
            System.out.println("Leap Year");  
        else  
            System.out.println("Not Leap Year");  
    }  
}  
...  

```

13. Count Vowels and Consonants

```
```java  
import java.util.*;

class Main{
 public static void main(String[] args){

 Scanner sc = new Scanner(System.in);
 String str = sc.nextLine().toLowerCase();

```

```

int vowels = 0, consonants = 0;
for(char ch : str.toCharArray()){
 if(Character.isLetter(ch)){
 if("aeiou".indexOf(ch) != -1)
 vowels++;
 else
 consonants++;
 }
}

System.out.println("Vowels: " + vowels);
System.out.println("Consonants: " + consonants);
}
}
...

```

## ## 14. Reverse a String

```

```java
import java.util.*;

class Main{

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String rev = "";

        for(int i = str.length() - 1; i >= 0; i--)
            rev += str.charAt(i);

        System.out.println(rev);

    }
}

```

```
}  
...  

```

15. Check Anagram

```
```java  
import java.util.*;

class Main{
 public static void main(String[] args){
 Scanner sc = new Scanner(System.in);

 String s1 = sc.nextLine();
 String s2 = sc.nextLine();

 char[] a = s1.toCharArray();
 char[] b = s2.toCharArray();

 Arrays.sort(a);
 Arrays.sort(b);

 System.out.println(Arrays.equals(a,b) ? "Anagram" : "Not Anagram");
 }
}
...

```

### **## 16. Remove Duplicates from String**

```
```java  
import java.util.*;  
  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  

```

```

String str = sc.nextLine();
String result = "";
for(char ch : str.toCharArray()){
    if(result.indexOf(ch) == -1)
        result += ch;
}
System.out.println(result);
}
}
...

```

17. Find Second Largest in Array

```

```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc = new Scanner(System.in);

 int n = sc.nextInt();

 int[] arr = new int[n];

 for(int i = 0; i < n; i++)

 arr[i] = sc.nextInt();

 int first = Integer.MIN_VALUE, second = Integer.MIN_VALUE;

 for(int num : arr){

 if(num > first){

 second = first;

 first = num;
 }
 }
 }
}

```

```

 } else if(num > second && num != first){
 second = num;
 }
}

System.out.println(second);
}
}
...

```

## ## 18. Linear Search

```

```java
import java.util.*;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i = 0; i < n; i++)
            arr[i] = sc.nextInt();
        int key = sc.nextInt();
        boolean found = false;
        for(int num : arr){
            if(num == key){
                found = true;
                break;
            }
        }
    }
}

```

```
System.out.println(found ? "Found" : "Not Found");
}
}
```
```

## ## 19. Binary Search (Sorted Array)

```
```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int[] arr = new int[n];

for(int i = 0; i < n; i++)

arr[i] = sc.nextInt();

int key = sc.nextInt();

int low = 0, high = n - 1;

boolean found = false;

while(low <= high){

int mid = (low + high) / 2;

if(arr[mid] == key){

found = true;

break;

}

else if(arr[mid] < key)

low = mid + 1;

else
```

```

high = mid - 1;
}
System.out.println(found ? "Found" : "Not Found");
}
}
...

```

20. Bubble Sort

```

```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int[] arr = new int[n];

for(int i = 0; i < n; i++)

arr[i] = sc.nextInt();

for(int i = 0; i < n-1; i++){

for(int j = 0; j < n-i-1; j++){

if(arr[j] > arr[j+1]){

int temp = arr[j];

arr[j] = arr[j+1];

arr[j+1] = temp;

}

}

}

for(int num : arr)

```

```
System.out.print(num + " ");
}
}
...
```

## ## 21. Selection Sort

```
```java  
import java.util.*;  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int[] arr = new int[n];  
        for(int i=0;i<n;i++) arr[i]=sc.nextInt();  
        for(int i=0;i<n-1;i++){  
            int minIndex=i;  
            for(int j=i+1;j<n;j++){  
                if(arr[j]<arr[minIndex])  
                    minIndex=j;  
            }  
            int temp=arr[i];  
            arr[i]=arr[minIndex];  
            arr[minIndex]=temp;  
        }  
        for(int num:arr)  
            System.out.print(num+" ");  
    }  
}
```

```
}  
...
```

22. Insertion Sort

```
```java  
import java.util.*;

class Main{
 public static void main(String[] args){
 Scanner sc=new Scanner(System.in);
 int n=sc.nextInt();
 int[] arr=new int[n];
 for(int i=0;i<n;i++) arr[i]=sc.nextInt();
 for(int i=1;i<n;i++){
 int key=arr[i];
 int j=i-1;
 while(j>=0 && arr[j]>key){
 arr[j+1]=arr[j];
 j--;
 }
 arr[j+1]=key;
 }
 for(int num:arr)
 System.out.print(num+" ");
 }
}
...
```

## ## 23. Matrix Addition

```
```java
import java.util.*;

class Main{

    public static void main(String[] args){

        Scanner sc=new Scanner(System.in);

        int r=sc.nextInt(), c=sc.nextInt();

        int[][] a=new int[r][c];

        int[][] b=new int[r][c];

        int[][] sum=new int[r][c];

        for(int i=0;i<r;i++)
            for(int j=0;j<c;j++)
                a[i][j]=sc.nextInt();

        for(int i=0;i<r;i++)
            for(int j=0;j<c;j++)
                b[i][j]=sc.nextInt();

        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                sum[i][j]=a[i][j]+b[i][j];

                System.out.print(sum[i][j]+" ");

            }

            System.out.println();

        }

    }

}

```
```

## ## 24. Transpose of Matrix

```
```java
import java.util.*;

class Main{

    public static void main(String[] args){

        Scanner sc=new Scanner(System.in);

        int r=sc.nextInt(), c=sc.nextInt();

        int[][] a=new int[r][c];

        for(int i=0;i<r;i++)

            for(int j=0;j<c;j++)

                a[i][j]=sc.nextInt();

        for(int j=0;j<c;j++){

            for(int i=0;i<r;i++){

                System.out.print(a[i][j]+" ");

            }

            System.out.println();

        }

    }

}

```
```

## ## 25. Count Frequency of Element in Array

```
```java
import java.util.*;

class Main{
```

```

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int[] arr=new int[n];
for(int i=0;i<n;i++) arr[i]=sc.nextInt();
int key=sc.nextInt();

int count=0;
for(int num:arr){
if(num==key) count++;
}
System.out.println(count);
}
}
...

```

26. Check if Array is Sorted

```

```java
import java.util.*;

class Main{

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int[] arr=new int[n];
for(int i=0;i<n;i++) arr[i]=sc.nextInt();
boolean sorted=true;
for(int i=0;i<n-1;i++){

```

```

if(arr[i]>arr[i+1]){
sorted=false;
break;
}
}
System.out.println(sorted ? "Sorted" : "Not Sorted");
}
}
...

```

## ## 27. Merge Two Arrays

```

``java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc=new Scanner(System.in);

int n1=sc.nextInt();

int[] a=new int[n1];

for(int i=0;i<n1;i++) a[i]=sc.nextInt();

int n2=sc.nextInt();

int[] b=new int[n2];

for(int i=0;i<n2;i++) b[i]=sc.nextInt();

int[] merged=new int[n1+n2];

for(int i=0;i<n1;i++)

merged[i]=a[i];

for(int i=0;i<n2;i++)

merged[n1+i]=b[i];

```

```
for(int num:merged)
System.out.print(num+" ");
}

}

...

```

## **## 28. Find Missing Number (1 to N)**

```
```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc=new Scanner(System.in);

int n=sc.nextInt();

int[] arr=new int[n-1];

for(int i=0;i<n-1;i++) arr[i]=sc.nextInt();

int total=n*(n+1)/2;

int sum=0;

for(int num:arr) sum+=num;

System.out.println(total-sum);

}

}

...

```

29. Count Words in a String

```
```java

```

```

import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc=new Scanner(System.in);

String str=sc.nextLine().trim();

if(str.isEmpty()){

System.out.println(0);

} else {

String[] words=str.split("\\s+");

System.out.println(words.length);

}

}

}

...

```

### **## 30. Remove All Spaces from String**

```

```java

import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc=new Scanner(System.in);

String str=sc.nextLine();

str=str.replaceAll(" ","");

System.out.println(str);

}

}

...

```

31. Find Duplicate Elements in Array

```
```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc = new Scanner(System.in);

 int n = sc.nextInt();

 int[] arr = new int[n];

 for(int i=0;i<n;i++) arr[i]=sc.nextInt();

 for(int i=0;i<n;i++){

 for(int j=i+1;j<n;j++){

 if(arr[i]==arr[j]){

 System.out.print(arr[i]+" ");

 break;

 }

 }

 }

 }

}

```
```

32. Move All Zeros to End

```
```java
import java.util.*;

class Main{
```

```

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int[] arr=new int[n];
for(int i=0;i<n;i++) arr[i]=sc.nextInt();
int index=0;
for(int i=0;i<n;i++){
if(arr[i]!=0){
arr[index++]=arr[i];
}
}
while(index<n){
arr[index++]=0;
}
for(int num:arr)
System.out.print(num+" ");
}
}
...

```

### **## 33. Rotate Array Right by 1 Position**

```

```java
import java.util.*;

class Main{

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();

```

```

int[] arr=new int[n];
for(int i=0;i<n;i++) arr[i]=sc.nextInt();
int last=arr[n-1];

for(int i=n-1;i>0;i--){
    arr[i]=arr[i-1];
}
arr[0]=last;
for(int num:arr)
    System.out.print(num+" ");
}
}
...

```

34. Check Palindrome String

```

```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc=new Scanner(System.in);

 String str=sc.nextLine();

 String rev="";

 for(int i=str.length()-1;i>=0;i--)

 rev+=str.charAt(i);

 System.out.println(str.equals(rev) ? "Palindrome" : "Not
 Palindrome");

 }
}

```

```
}
...

```

### **## 35. Count Number of Digits**

```
```java  
import java.util.*;  
  
class Main{  
    public static void main(String[] args){  
        Scanner sc=new Scanner(System.in);  
        int n=sc.nextInt();  
        int count=0;  
        while(n!=0){  
            n/=10;  
            count++;  
        }  
        System.out.println(count);  
    }  
}  
...  

```

36. Sum of Elements in Array

```
```java  
import java.util.*;

class Main{
 public static void main(String[] args){
 Scanner sc=new Scanner(System.in);

```

```

int n=sc.nextInt();
int[] arr=new int[n];
int sum=0;
for(int i=0;i<n;i++){

arr[i]=sc.nextInt();
sum+=arr[i];
}
System.out.println(sum);
}
}
...

```

### **## 37. Find Minimum Element in Array**

```

```java
import java.util.*;

class Main{

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int[] arr=new int[n];
for(int i=0;i<n;i++) arr[i]=sc.nextInt();
int min=arr[0];
for(int i=1;i<n;i++){
if(arr[i]<min)
min=arr[i];
}
}
}

```

```
System.out.println(min);  
}  
}  
``
```

38. Pattern Printing (Right Triangle)

Input: 4

Output:

```
*  
  
**  
  
***  
  
****
```

```
``java
```

```
import java.util.*;  
  
class Main{  
  
    public static void main(String[] args){  
  
        Scanner sc=new Scanner(System.in);  
  
        int n=sc.nextInt();  
  
        for(int i=1;i<=n;i++){  
  
            for(int j=1;j<=i;j++){  
  
                System.out.print("*");  
  
            }  
  
            System.out.println();  
  
        }  
  
    }  
  
}
```

```

### **## 39. Power of a Number**

```
```java
import java.util.*;

class Main{

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int base=sc.nextInt();
        int exp=sc.nextInt();
        int result=1;
        for(int i=1;i<=exp;i++)
            result*=base;
        System.out.println(result);
    }
}
```
```

### **## 40. Decimal to Binary**

```
```java
import java.util.*;

class Main{

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
```

```

String binary="";
while(n>0){
    binary=(n%2)+binary;
    n/=2;
}
System.out.println(binary);
}
}
...

```

41. Binary to Decimal

```

```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc = new Scanner(System.in);

 String binary = sc.next();

 int decimal = 0;

 int power = 0;

 for(int i = binary.length()-1; i >= 0; i--){

 if(binary.charAt(i) == '1'){

 decimal += Math.pow(2, power);

 }

 power++;

 }

 System.out.println(decimal);

 }

}

```

```
}
...
```

### ## 42. Check Perfect Number

```
```java  
import java.util.*;  
  
class Main{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int sum = 0;  
        for(int i = 1; i <= n/2; i++){  
            if(n % i == 0)  
                sum += i;  
        }  
        System.out.println(sum == n ? "Perfect" : "Not Perfect");  
    }  
}  
...
```

43. Strong Number

```
```java  
import java.util.*;

class Main{
 public static int factorial(int n){
```

```

int fact = 1;
for(int i = 1; i <= n; i++)
 fact *= i;
return fact;
}

public static void main(String[] args){
 Scanner sc = new Scanner(System.in);
 int n = sc.nextInt();
 int temp = n, sum = 0;
 while(n != 0){
 int digit = n % 10;
 sum += factorial(digit);
 n /= 10;
 }
 System.out.println(temp == sum ? "Strong" : "Not Strong");
}
}
```

```

44. Count Even and Odd Numbers in Array

```

```java
import java.util.*;

class Main{

 public static void main(String[] args){
 Scanner sc = new Scanner(System.in);
 int n = sc.nextInt();
 int[] arr = new int[n];
 }
}
```

```

```

int even = 0, odd = 0;

for(int i=0;i<n;i++){
arr[i]=sc.nextInt();
if(arr[i]%2==0)
even++;

else
odd++;
}

System.out.println("Even: " + even);
System.out.println("Odd: " + odd);
}
}
...

```

45. Find Intersection of Two Arrays

```

```java
import java.util.*;

class Main{

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n1=sc.nextInt();
int[] a=new int[n1];
for(int i=0;i<n1;i++) a[i]=sc.nextInt();
int n2=sc.nextInt();
int[] b=new int[n2];
for(int i=0;i<n2;i++) b[i]=sc.nextInt();

```

```
for(int i=0;i<n1;i++){
 for(int j=0;j<n2;j++){
 if(a[i]==b[j]){
 System.out.print(a[i]+" ");
 break;
 }
 }
}
```

## ## 46. Check Substring

```
```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc=new Scanner(System.in);

String str=sc.nextLine();

String sub=sc.nextLine();

if(str.contains(sub))

System.out.println("Substring Present");

else

System.out.println("Substring Not Present");

}

}

```
```

## **## 47. Remove Specific Character from String**

```
```java
import java.util.*;

class Main{

public static void main(String[] args){

Scanner sc=new Scanner(System.in);

String str=sc.nextLine();

char ch=sc.next().charAt(0);

str=str.replace(Character.toString(ch),"");

System.out.println(str);

}

}

...

```

48. Sum of Prime Numbers up to N

```
```java
import java.util.*;

class Main{

public static boolean isPrime(int n){

if(n<=1) return false;

for(int i=2;i<=Math.sqrt(n);i++){

if(n%i==0) return false;

}

return true;

}

}

```

```

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int sum=0;
for(int i=2;i<=n;i++){
if(isPrime(i))
sum+=i;
}
System.out.println(sum);
}
}
...

```

#### **## 49. Reverse Words in a Sentence**

```

```java
import java.util.*;

class Main{

public static void main(String[] args){
Scanner sc=new Scanner(System.in);
String str=sc.nextLine();
String[] words=str.split(" ");
for(int i=words.length-1;i>=0;i--){
System.out.print(words[i]+" ");
}
}
}
}
...

```

50. Two Sum Problem

```
```java
import java.util.*;

class Main{

 public static void main(String[] args){

 Scanner sc=new Scanner(System.in);

 int n=sc.nextInt();

 int[] arr=new int[n];

 for(int i=0;i<n;i++) arr[i]=sc.nextInt();

 int target=sc.nextInt();

 for(int i=0;i<n;i++){

 for(int j=i+1;j<n;j++){

 if(arr[i]+arr[j]==target){

 System.out.println(i+" "+j);

 return;

 }

 }

 }

 System.out.println("No Pair Found");

 }

}

```
```